

The Complete Book 3 Story, Formulas & Traps

All 24 Readings, All 3 Study Sessions:

Operational Risk Overview (40–46) · Operational Risk Focus Areas (47–56) · Capital & Regulatory Frameworks (57–63)

Built directly from your Schweser Book 3 — every formula, every reading's story, worked examples, and the exam traps GARP likes to test — threaded into one continuous narrative from Reading 40 to Reading 63. Complete companion guide for Operational Risk and Resilience.

Study Session 7 — Operational Risk Overview: Readings 40–46. The ORM framework spine: what op risk is, how it's governed, identified, measured, mitigated, reported, and integrated into ERM.

Study Session 8 — Operational Risk Focus Areas: Readings 47–56. Named risk domains and case studies: cyber, information security, AML/financial crime, outsourcing/third-party risk, investor protection, model risk, and stress testing.

Study Session 9 — Capital and Regulatory Frameworks: Readings 57–63. The quant core: RAROC, economic capital, capital planning, and the full arc of Basel I → II → 2.5 → III → the finalized SMA operational risk capital formula.

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STUDY SESSION 7 OF 9

Operational Risk Overview

This session builds the operational risk management (ORM) framework end to end, in the exact order a risk manager would use it: define and categorize the risk (R40), govern it (R41), identify it (R42), measure and assess it (R43 — the one quant-heavy reading here), mitigate it (R44), report it (R45), and integrate it into enterprise-wide risk management and stress testing (R46). Almost everything here is conceptual and definitional — the exam rewards precise recall of frameworks (three lines of defense, RCSA, KRI/KPI/KCI, the Basel event-type taxonomy) over calculation.

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Introduction to Operational Risk and Resilience

GARP, *Operational Risk and Resilience*, Ch. 1

The story

Operational risk is defined by exclusion as much as by inclusion: it's the risk of loss from *failed processes, people, systems, and external events* — everything that isn't market or credit risk, but still hits the P&L. This reading gives you the classification skeleton (the seven Basel event-type categories) and the characteristics that make op risk uniquely hard to manage compared to market/credit risk, then closes with operational **resilience** — the newer idea that some disruptions can't be prevented, only survived.

The seven Basel II event-type categories (memorize cold)

Category	Example
Internal fraud	Employee misappropriation, intentional mismarking of positions
External fraud	Robbery, hacking, third-party theft
Employment practices & workplace safety	Discrimination claims, workers' comp
Clients, products & business practices	Fiduciary breaches, misselling, antitrust
Damage to physical assets	Natural disasters, terrorism
Business disruption & system failures	Hardware/software failures, utility outages
Execution, delivery & process management	Data entry errors, failed mandatory reporting

TRAP

Execution/delivery/process management (EDPM) events are high-frequency, low-severity; clients/products/business practices (CPBP) events are low-frequency, high-severity. The exam likes to test this frequency/severity contrast directly (also seen again in R42's taxonomy discussion).

Five characteristics that make op risk hard to manage

Heterogeneous (wildly different event types) · idiosyncratic (firm-specific, not market-driven) · heavy-tailed (rare catastrophic events dominate) · interconnected (one failure cascades into others) · dynamic (risk profile shifts as technology/business models change).

Operational resilience — the newer add-on

Resilience shifts the mindset from "prevent all disruptions" to "assume severe, infrequent disruptions will happen — make sure the firm survives them." Elements: business continuity, key (important) services, impact tolerance levels, disruption processes, feedback. The BCBS's seven principles of operational resilience: governance, operational risk management, business continuity planning and testing, mapping interconnections and interdependencies, third-party dependency management, incident management, and ICT/cybersecurity.

THREAD

"Important business services" and "impact tolerances" reappear with much more mechanical detail in R43's operational resilience section (the seven-step process) — this reading gives you the vocabulary, R43 gives you the process.

Risk Governance

GARP, Operational Risk and Resilience, Ch. 2

The story

R41 answers "who's in charge of operational risk?" at three levels: the regulatory framework (Basel II's three pillars, applied here specifically to op risk), the committee structure (bottom → middle → top), and the three lines of defense — the single most-recycled framework in this entire book, first seen in Credit Risk Book 2 and now given its fullest treatment here.

Basel II's three pillars, applied to operational risk

Pillar 1 (regulatory capital): minimum capital for unexpected op-risk losses, minimum liquidity coverage, BCBS's Principles for Sound Management of Operational Risk. **Pillar 2** (supervisory review): extra capital for risks not in Pillar 1 (concentration, compliance, governance risk), plus voluntary disclosure/self-assessment. **Pillar 3** (market discipline): mandatory quarterly/annual risk disclosures.

Committee structure: bottom → middle → top

Lowest: numerous small committees per business activity/country, feeding data upward. Middle: organization risk committee aggregates firm-wide op risk, reports to the top. Top: board (enterprise) risk committee oversees everything, evaluates major incidents, and must have members with current, relevant risk management experience.

Three lines of defense — the full mechanics

Line	Who	Key duty
1 (+ "1.5")	Risk owners (front line); sometimes a risk champion bridges to Line 2	Generate, measure, and manage their own risk
2	Independent risk management	Develop ORM policy, cross-examine Line 1's work, must have zero involvement in Line 1 to avoid self-review threat
3	Internal audit	Fully independent review of Lines 1 and 2; keeps its own risk list separate from Line 2's

TRAP

Line 2 must have *zero* involvement in Line 1 work to avoid a "self-review threat" — a subtle but frequently tested independence requirement. Also: some areas (legal/ compliance, IT security) genuinely straddle multiple lines and can't be cleanly classified into just one.

Risk appetite and risk culture

The board sets risk appetite: evaluate significant risks, set limits distinguishing acceptable from unacceptable outcomes, establish controls. Risk appetite statements should be simple, explain *why* certain risks are accepted/declined/minimized, and (increasingly) include disruption/impact tolerances for key services. Risk culture is linked to ethics ("tone at the top"), a board-level code of conduct, and compensation structure that doesn't reward excessive risk-taking.

Risk Identification

GARP, Operational Risk and Resilience

The story

Before you can measure or mitigate a risk, you have to find it. R42 gives you two directions to look (top-down from the board, bottom-up from business units), the scenario-analysis machinery for extreme-but-plausible events, and the taxonomy that structures everything — the same Basel seven-category system from R40, now one level deeper.

Top-down vs. bottom-up identification

Top-down	Bottom-up
Analyzing exposures & vulnerabilities	Event and loss data analysis
Risk wheel	Risk and control self-assessment (RCSA)
Analyzing emerging risks	Process mapping

Combining both gives a more complete picture than either alone — top-down catches strategic/emerging risks the business units might miss; bottom-up catches operational detail senior management can't see.

Scenario analysis & scenario workshops

Scenarios are extreme, financially severe, plausible-but-unlikely events used to bridge risk identification into quantitative assessment. Neutrality is critical (regulators want consistent results across iterations) — achieved through empirical evidence and documented rationale for each scenario. Scenario workshops run in two phases: **generation** (brainstorm broadly, then screen) and **selection** (combine, drop, or add scenarios to reach a final list for assessment).

The Basel taxonomy, one level deeper

Three levels: Level 1 = the seven broad event categories (from R40); Level 2 = 20 more specific categories; Level 3 = concrete examples. EDPM losses: small dollar amount, very high frequency. CPBP losses: infrequent, very large. Many banks build their own tailored taxonomy on top of Basel's (accounting for size, regulation, geography) — the **ORX taxonomy** (2019) is a well-known alternative, expanding from 7 to 14 Level-1 categories by promoting some Basel Level-2 risks (third-party failure, statutory reporting/tax, business continuity, data management, information security/cyber, and model risk) up to Level 1 status.

TRAP

The ORX taxonomy elevating "model risk" and "information security/ cyber risk" to Level-1 status (rather than burying them as Level-2 sub-items under a broader category) reflects exactly why R47–48 (cyber) and R54–55 (model risk) get entire dedicated readings later in this book — these risks earned their own spotlight because they didn't fit neatly in the original 7-category Basel scheme.

Risk Measurement and Assessment

The story

This is the quant heart of Session 7 — the reading where op risk gets numbers attached. Three big ideas: qualitative assessment (RCSAs, heatmaps), quantitative assessment (fault tree analysis, the FAIR model), and the loss distribution approach (LDA) that underlies operational risk capital modeling. Also closes with the seven-step operational resilience process teased in R40.

Operational loss data — regulatory requirements

BCBS minimums: ≥10 years of data, minimum collection threshold of €20,000, internal losses classified to Basel event types, occurrence and recovery dates reported. Four relevant dates per event: occurrence → discovery → reporting → accounting (settlement can lag occurrence by years). **Boundary events** occur in one risk class but are caused by another (e.g., a credit loss caused by human/operational error) — boundary events tied to credit risk are treated as credit risk (already in RWA, not double-counted in op risk); boundary events tied to market risk stay in operational losses.

RCSAs, KRIs, KPIs, KCIs

RCSA examines inherent risk (no controls) and residual risk (after controls), is primarily qualitative (judgment-based, not meant to quantify exposures), and typically runs annually. Backtesting RCSAs against actual incidents tends to reveal *underestimated likelihood* and *overestimated severity*. Heatmaps combine a likelihood scale (remote/unlikely/possible/likely/highly likely) with an impact scale (low/medium/high/very high/ extreme) into green/yellow/amber/red — but these are qualitative labels, not numbers to be multiplied together.

Indicator	Measures	Example
KRI	Exposure level (rise/fall in likelihood or impact)	Transactions processed per employee
KPI	How effectively the bank operates	Customer complaint count, IT downtime
KCI	How effectively controls operate	Overdue business continuity plan reviews

Fault tree analysis (FTA) — the calculation trap

Independent control failures multiply together for a joint probability (AND conditions).

WORKED EXAMPLE (FROM THE MODULE QUIZ)

Phishing received 99% × firewall fails 5% × employee fails to catch it 10% × detection fails 3% × information exits 1% = $0.99 \times 0.05 \times 0.10 \times 0.03 \times 0.01 = 0.0001485\%$.

In reality, controls are rarely fully independent (they're often designed together) — conditional (Bayesian) probabilities give a more realistic estimate. Scaling matters too: a tiny per-event probability times a large number of employees/transactions can still produce a material aggregate likelihood.

FAIR model & the Swiss cheese model

FAIR (Factor Analysis of Information Risk): (1) determine risk factors and interrelations, (2) measure each factor, (3) computationally combine them (often via Monte Carlo) into a loss distribution. **Swiss cheese model**: each control layer has holes; bad outcomes happen only when holes in multiple layers momentarily align. Defense: keep control failure rates uncorrelated and maintain compensating controls.

Loss distribution approach (LDA)

Splits losses into **frequency** (discrete, usually Poisson — one parameter λ = both mean and variance) and **severity** (continuous, fat-tailed, usually lognormal), modeled independently then convoluted (often via Monte Carlo) into an aggregate loss distribution. The **99.9th percentile** of that aggregate distribution is the stand-alone capital requirement under LDA. Units of measure (UoM) should be homogeneous (similar loss drivers) but homogeneity trades off against data volume — finer UoMs mean less data per bucket. Copulas aggregate across UoMs, capturing tail dependence.

Extreme value theory (EVT)

Method	Approach
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Block maxima (Fisher-Tippett)	Max loss per equally-spaced period per UoM
Peaks-over-threshold (POT)	Losses above a high threshold, fit with the generalized Pareto distribution (GPD)

EVT only really works when there's one dominant underlying cause generating losses — a poor fit for operational risk's genuinely heterogeneous loss drivers, which limits its practical usefulness here.

Combining internal and external loss data

Scaling (size/inflation adjustments) · cut-off mix (blend in external data above a threshold where internal data thins out) · filtering (explicit inclusion/exclusion rules to prevent manipulation).

Operational resilience — the seven steps

1. Determine important business services (IBSs)
2. Establish impact tolerances for each IBS
3. Map IBSs and the resources needed to deliver them
4. Design harsh-but-realistic scenarios to test vulnerabilities
5. If tolerances are breached, review lessons learned and remediate
6. Ensure communication plans are ready to execute
7. Perform an annual board-approved self-assessment

Business impact analysis (BIA) underpins step 2. Single points of failure (SPOFs) — e.g., a key employee with undocumented specialist knowledge, or a supplier with no substitute — are treated as resilience KRIs.

TRAP

Resilience is *not* the same as business continuity management (BCM). BCM covers every business process's continuity/recovery broadly; resilience is narrower — it applies specifically to "important business services" under non-extreme conditions and impact tolerances. "Intolerable impacts" that breach tolerances fall outside resilience's scope by definition.

Risk Mitigation

The story

Once risk is measured, what do you actually do about it? R44 gives the four generic responses, the four types of internal controls, the human-error taxonomy behind control design, and the practical playbook for crisis response (BCM/DRP) and risk transfer (insurance, outsourcing, reputation management).

Four T's of risk response

Tolerate (fully accept) · **Treat** (mitigate impact via action/remedy) · **Transfer** (insurance, third parties) · **Terminate** (eliminate exposure by discontinuing the activity).

Four types of internal controls

Type	Function
Preventive	Lower the chance of an event happening at all
Detective	Alarm/flag the event quickly to limit loss
Corrective	Reduce negative impact after the fact (BCP, backups)
Directive	Guidance for performing a task correctly (policies, training)

Automated controls beat manual ones for reliability but bring their own risks: Type 1 (false positive) and Type 2 (false negative) errors, and outages from system downtime/overcapacity. Weak-control patterns to watch for: "optimistic" controls (too dependent on one individual, superficial) and "collective" controls (diffused responsibility reduces accountability) — and the common bad habit of just piling on more of the same control type after a failure instead of diagnosing the actual gap.

Human error taxonomy (drives control design)

Type	Voluntary?	Cause	Remedy
Slip	No	Inattention/distraction	Better workspace design, clearer accountability
Rule-based mistake	Yes	Flawed rule followed correctly	Improve the rule, strengthen controls
Knowledge-based mistake	Yes	Wrong choice in an unfamiliar situation	Better documentation, training, escalation paths
Violation	Yes (not error)	Deliberately breaking a known rule	Better supervision/detection (cameras, call recording)

Lean Six Sigma (DMAIC: define, measure, analyze, improve, control) and quality improvement (PDSA: plan, do, study, act) are the two named process-improvement cycles for reducing operational error at the design stage.

Contingency planning & crisis response

BCM (business continuity management) is the action plan for crisis scenarios, part of the broader BCP. Success factors: speed, competence (right specialist per task), transparency. Four phases of a major op-risk event: crisis → emergency response → recovery → restoration.

Insurance, outsourcing, reputation

Insurance trades an ongoing premium for reduced loss volatility — many large banks self-insure small losses since insurance costs can exceed the benefit at that scale. Outsourcing transfers risk to a third party, but the degree of actual risk transfer is case-specific — sometimes it's genuine transfer, sometimes really risk-sharing. Reputation management splits into prevention (image/relationship building, contingency planning) and mitigation (fast, transparent communication).

Risk Reporting

The story

R45 closes the loop: identify (R42) → measure (R43) → mitigate (R44) → **report**, so the right people at the right altitude get the right information to act on. The organizing device is the "reporting cake" — three tiers matching report detail to audience seniority.

The reporting cake

Tier	Audience	Content
1 (bottom)	Business line managers, risk champions	Full metric set, monitored daily
2 (middle)	Operational risk committee	Action-required items only; summaries monthly/quarterly
3 (top)	Executive & board risk committee	High-level only

A comprehensive internal ORM report includes: prioritized risks (top-10 + outlook), heatmap/risk register, risk appetite metrics, KRIs and issue monitoring, risk events/near misses, action plans/remediation, and emerging risks/horizon scanning.

Reporting challenges (why op risk reporting is genuinely hard)

Event data asymmetry: few high-severity/low-frequency events dominate, distorting simple averages. **Escalation:** large events and significant near-misses need explicit upward escalation. **Small frequent losses:** individually immaterial but worth analyzing for structural control gaps. **Distribution shape:** op risk losses aren't symmetric/bell-shaped — median/quartiles beat means. **Qualitative data aggregation:** three options — conversion and addition, categorization, or worst-case reporting.

External reporting (Pillar 3) & combined assurance

Pillar 3 disclosure covers qualitative information, 10 years of historical losses, and the business indicator and subcomponents (the exact BI machinery detailed later in R63). Regulatory notification triggers: reputation, resilience, materiality, and stability criteria. **Combined assurance** aligns all three lines of defense so senior management/audit committee get one coherent picture rather than three overlapping ones.

Integrated Risk Management

The story

R46 zooms out from operational risk specifically to enterprise risk management (ERM) generally, then zooms back in on stress testing as the connective tissue between ERM and the capital readings coming in Session 9. Notice the direct callback to RAROC (fully developed in R57) and economic capital (R58) — this reading previews both.

ERM's governance-culture-appetite triad (+ capital/stress testing for financial firms)

Governance sets roles/responsibilities across the three lines of defense (plus external audit as an informal fourth line, and a Risk Committee overseeing everything). Culture is the behaviors/values around risk. Appetite defines tolerance levels. For financial firms specifically, risk capital and stress testing form a fourth pillar.

RAROC and economic capital, previewed

Economic capital $\approx$ combined Pillar 1 + Pillar 2 requirements — the funds needed to cover unexpected losses. RAROC = expected after-tax risk-adjusted net income / economic capital (full mechanics in R57). Capital needs aggregate across risk classes at less than the simple sum (assuming imperfect correlation) — operational risk in particular offers meaningful diversification benefit due to its low correlation with market/credit risk.

Stress testing taxonomy

Two dimensions: analytical approach (quantitative vs. qualitative) and risk type (measurable vs. immeasurable). Three approaches: **parameter (model) testing** (shock parameter values to test model robustness), **macroeconomic testing** (shock scenarios to test solvency/resilience), **reverse testing** (start from a failure outcome, work backward to the circumstances that would cause it).

A comprehensive operational risk stress-testing framework has three modules: expected nonlegal loss forecast, legal loss, and idiosyncratic scenario add-on. Frequency/severity modeling uses regression against macro variables, or a (modified/conditional) LDA with Monte Carlo — a conditional LDA is a middle ground between full regression and plain LDA. Severity is harder to model credibly than frequency.

STUDY SESSION 8 OF 9

Operational Risk Focus Areas

Ten readings covering named risk domains and real case studies — the most memorization-heavy, least formula-heavy session in the book. Cyber (R47–48), financial crime and AML (R49–50), third-party/ outsourcing risk (R51–52), investor protection (R53), model risk (R54–55), and stress testing at banks (R56). The case-study readings (48, 50, 52, 53, 55) are short and pair a named real-world failure with the lessons GARP wants you to draw from it — fast reads, but the specific failure-pattern (which control failed, why) is exactly what gets tested.

47 Cyber-Resilience · 48 Cyberthreats Case Study · 49 AML/CFT · 50 Financial Crime Case Study · 51 Outsourcing Risk · 52 Third-Party Risk Case Study · 53 Investor Protection Case Study · 54 Model Risk Management · 55 Model Risk Case Study · 56 Stress Testing Banks

Cyber-Resilience: Range of Practices

The story

A survey of how regulators globally approach cyber risk — deliberately non-prescriptive (practices vary a lot by jurisdiction), so the exam tests the *categories* of practice rather than one "correct" standard.

Five areas of cyber governance

Cybersecurity strategy · management roles & responsibilities (board/senior management, often the three-lines model again) · awareness (training, risk culture) · architecture & standards · workforce (IT staff training, defined roles).

Four areas of cyber risk management & response

Supervision of cyber-resilience · information security controls (threat detection, business impact analysis, recovery planning) · incident response and recovery · cybersecurity/resilience metrics (no standardized quantitative metrics exist yet — forward-looking indicators are increasingly valued over backward-looking data).

Information-sharing channels (know all five)

Bank-to-bank (depends on trust level) · bank-to-regulator (usually mandated) · regulator-to-regulator (rare, cross-border) · regulator-to-bank (uncommon, little formal guidance) · banks/regulators-to-security agencies (mostly voluntary).

Third-party cyber risk dimensions

Governance of interconnections · business continuity/availability · information confidentiality & integrity · visibility into third-party (and sub-outsourcing) relationships · auditing/testing rights · resources and skills of the outsourcing institution's own oversight staff.

Case Study: Cyberthreats and Information Security Risks

The story

Pairs a framework overview with the 2017 Equifax breach as the illustrative failure — the exam wants you to connect specific control gaps to the generic frameworks named here.

Three cybersecurity frameworks

Framework	Focus
NIST CSF	Voluntary; five steps — identify, protect, detect, respond, recover
CIS Critical Security Controls	Detailed, continuously updated technical control checklist
ISO 27001	Certifiable ISMS (Information Security Management System) standard

Core protection triad: **confidentiality, integrity, availability (CIA)**. Controls split into behavioral (training, awareness, password hygiene) and technical (prevention/detection/mitigation tooling).

Equifax (2017) — five failure points

Insufficient IT asset inventory · failure to enforce patch management policy · inconsistent employee communication · an expired SSL certificate · poor external/public communication.

TRAP

Equifax's core failure wasn't a lack of a framework — it was failing to *execute* a known patch management policy. The exam likes to test that having a policy on paper (directive control) isn't the same as an operating (preventive/detective) control.

Sound Management of Risks Related to Money Laundering and Financing of Terrorism

The story

AML/CFT is the three-lines-of-defense model applied to a specific problem: knowing your customer well enough to stop criminal money from flowing through the bank. The reading layers a cross-border dimension on top for global banking groups.

Three lines of defense for AML/CFT

Line 1: front-office/customer-facing units — identify, assess, control ML/FT risk via written, communicated policies. Line 2: chief AML/CFT officer — ongoing monitoring. Line 3: internal audit of AML/CFT policy adherence.

Customer risk factors & identification (FATF standards)

Background, occupation, source of income/wealth, country of origin/residence, product/service usage pattern, account purpose, linked accounts. Identification/verification documentation: passports, ID cards, driving licenses, transaction records, business correspondence. Banks must also identify/verify persons acting on behalf of beneficial owners.

Cross-border groups

Integrate customer/beneficial-owner/fund information group-wide · monitor significant relationships on a consolidated basis regardless of on/off-balance-sheet treatment · appoint a group-wide chief AML/CFT officer · coordinate group-wide information sharing (subject to local data-protection law) · head office must be informed about high-risk customers. Supervisors follow FATF Recommendation 26 and the Core Principles for Effective Banking Supervision, applying a risk-based approach.

Case Study: Financial Crime and Fraud

The story

Builds out the fraud/AML vocabulary from R49 with precise definitions, then anchors it to the 2022 USAA Federal Savings Bank fine — notable because the penalty was for *weak controls*, not proven theft or loss.

Internal vs. external fraud (precise definitions)

Internal fraud: losses from acts intending to defraud/misappropriate/circumvent rules involving at least one internal party (excludes discrimination events).

External fraud: same intent, but by a third party only (robbery, forgery, hacking).

Money laundering's three phases

Placement (introduce illicit funds into the financial system) → **layering** (obscure the funds' origin through complex transactions) → **integration** (reintroduce the now-"clean" funds into the legitimate economy).

Four-control framework (shared by internal fraud, external fraud, and AML)

Selection (incl. know-your-customer/KYC due diligence for AML) → prevention → detection → deterrents.

USAA FSB (March 2022)

Fined for failing to meet Bank Secrecy Act (BSA)/AML minimum standards and inadequate suspicious-activity monitoring/reporting — *not* for a specific known theft. Following sanctions, firms typically run AML remediation programs ("lookbacks") reviewing client information and re-reporting suspicious activity.

TRAP

The USAA fine is a clean example that regulatory penalties can follow from *control deficiencies alone* — no proven loss event is required. Don't assume a regulatory fine implies an actual quantifiable loss occurred.

Guidance on Managing Outsourcing Risk

The story

The regulatory-guidance counterpart to R52's real-world case study — six risk categories, six program elements, and the specific contract clauses regulators expect to see.

Six risks from outsourcing

Compliance · concentration · reputation · country · operational · legal.

Six-part outsourcing risk management program

Risk assessments · due diligence in selecting providers · contract provisions · incentive compensation review · oversight/monitoring · business continuity/contingency plans.

Due diligence — three review areas

Business background, reputation, and strategy · financial performance and condition · operations and internal controls.

Contract provisions (a long, testable list)

Scope · cost/compensation · incentive compensation · right to audit · performance standards · oversight/monitoring · confidentiality/security · ownership and license · indemnification · default/termination · dispute resolution · limits on liability · insurance · customer complaints · provider's own business resumption plan · foreign-based provider considerations · subcontracting rights.

Case Study: Third-Party Risk Management

The story

Extends R51's outsourcing framework into full third-party risk management (TPRM) — including the less obvious fourth- and fifth-party exposures (your vendor's vendors) — then anchors it with Capital One and Morgan Stanley failures whose shared lesson is the reading's single most important takeaway.

TPRM lifecycle (five stages)

1. Business model decision — what should even be outsourced
2. Evaluation, risk rating, due diligence — proportional to relationship complexity/length
3. Contracts, SLAs, contract management — periodically reviewed
4. Ongoing monitoring — reassessment triggers defined in advance
5. Remediation or termination — grievance period, exit/termination clause

TRAP

The single lesson GARP wants from both Capital One and Morgan Stanley: **accountability cannot be outsourced**, even when the underlying function can be. A vendor's failure is still the bank's regulatory and reputational problem.

Case Study: Investor Protection and Compliance Risks in Investment Activities

The story

Four named regulations, then three bank fines (UBS, JPMorgan, Deutsche Bank) illustrating that penalties are sized to be punitive enough to exceed any benefit the firm gained from the violation — deterrence, not just restitution.

Four key regulations

Regulation	Focus
MiFID (2007, EU)	Business/organizational conduct, disclosure & reporting to prevent market abuse
MiFID II / MiFIR (2014)	Advisor pay, conflicts of interest, best execution, independent advice, product governance
Investor Protection Act (part of Dodd-Frank, 2009, US)	Whistleblower protection, stronger OTC derivatives rules, SEC oversight expansion
Volcker Rule (Dodd-Frank)	Prohibits proprietary trading by commercial banks; also created the CFPB

Regulatory compliance splits into two categories: (1) suitability, disclosure, and fiduciary responsibilities; (2) improper business/market practices. Effective compliance risk management needs strong trade/employee supervision, robust middle/back-office functions, training, and ethics culture.

Supervisory Guidance on Model Risk Management

The story

Model risk is a distinct animal from other operational risks: it isn't a single event, it's a systematic risk baked into every decision the model informs. Two ways it bites: the model itself is wrong (errors), or a correct model is used the wrong way (misuse/out-of-context application) — a distinction that repeats in R55's case studies.

Sizing model risk

Depends on model complexity, uncertainty in inputs/assumptions, and potential impact on users. Mitigation: restrict use, analyze performance, continually recalibrate, and always contextualize model output against other information rather than treating it as gospel.

Model development discipline

Clear objective aligned with intended use, thorough documentation of strengths/weaknesses, rigorous testing (both quantitative and qualitative aspects — don't neglect the qualitative side).

Three elements of model validation

Element	What it checks
Conceptual soundness	Documentation review, live testing, single/multi-variable sensitivity analysis
Ongoing monitoring	Process verification (data input integrity) + benchmarking (compare vs. other models/data)
Outcomes analysis	Parallel analysis of amended models + backtesting (out-of-sample period, consistent with the model's forecast horizon)

TRAP

Backtesting for model validation must use a time period *different from* the one used to build the model, but *consistent with* the model's actual forecast horizon — both conditions are tested, not just one.

Case Study: Model Risk and Model Validation

The story

Puts numbers and tiers around R54's abstractions: a formal model risk tiering system, and three concrete failure modes drawn from real cases.

Model risk tiering (drives how strict validation must be)

Materiality of output · model complexity · whether it's client-facing · whether it's used for regulatory compliance. The model risk management (MRM) team sets documentation, data-quality, and version-control standards proportional to the tier.

Three failure modes (map directly onto case examples)

Failure mode	What goes wrong
Invalid assumptions	The model's foundational premises don't hold in the real world
Implementation error	The model logic is coded/executed incorrectly
Input measurement error	The data feeding the model is wrong or mismeasured

THREAD

These three failure modes are just R54's "execution risk vs. conceptual error" split, cut a slightly different way — invalid assumptions and implementation error are both "conceptual/execution" failures, while input measurement error is closer to a data-quality (R43-style loss-data) problem wearing a model-risk label.

Stress Testing Banks

The story

Closes Session 8 by making the case for supervisory stress testing as a post-crisis risk tool superior to static capital ratios, then digs into why designing a genuinely *coherent* stress test is hard, and how macro shocks get translated into bank-specific losses — direct setup for R59's capital planning and the Basel capital math in R60–63.

The disclosure evolution: SCAP → CCAR

Program	Year	Disclosure level
SCAP (US)	2009	Full bank-level disclosure — high transparency, first post-crisis macro-prudential test
CCAR (initial)	2011	Only macro-scenario results published, not bank-level
CCAR (revised)	2012	Back to near-SCAP level of bank-level detail

EBA's Irish and European stress tests required significant disclosure too — the goal in every case was restoring market trust in the banking system through transparency.

The coherence problem

Scenarios must be extreme *and* plausible simultaneously — genuinely hard because real risk is multi-factor (a scenario extreme in one dimension may be unrealistic when all dimensions are considered jointly).

Macro-to-micro translation

Stress tests start from macro factors (unemployment, GDP growth, home price index) and must map them down to bank-specific, product-specific, geography-specific revenue and loss outcomes — the same "intermediate risk factor" mapping problem that stress-testing frameworks in R46 and R58 also wrestle with. Mechanically: the starting balance sheet generates quarter 1's stressed income/loss, which produces the quarter-end balance sheet, feeding into quarter 2, and so on — typically over a two-year horizon, with capital and liquidity ratios required to hold throughout.

STUDY SESSION 9 OF 9

Capital and Regulatory Frameworks

The quant payoff of the book: RAROC (R57) and economic capital (R58) give you the firm-level capital allocation logic; R59 shows how a real bank holding company operationalizes it; and R60–63 walk the entire regulatory arc — Basel I's risk-weighting innovation, Basel II's three pillars and IRB formula, the post-crisis Basel 2.5/III overhaul (stressed VaR, capital buffers, liquidity ratios, CoCos), and finally the 2017 reforms culminating in the standardized operational risk capital formula (SMA) in R63. This is the most formula-dense session in the book — budget real drilling time here.

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Risk Capital Attribution and Risk-Adjusted Performance Measurement

The story

RAROC answers one question with real teeth: is this loan, project, or business unit actually worth the capital it ties up? This reading builds RAROC from the ground up (basic formula → detailed capital-budgeting formula → hurdle rate comparison → a systematic-risk-adjusted version) and closes with the genuinely hard part: diversification benefits are real, but allocating them fairly across business units is contested and political.

Risk capital vs. economic capital vs. regulatory capital

DEFINITION

economic capital = risk capital + strategic risk capital

Regulatory capital differs from risk capital in three ways: it only applies to regulated industries; it's set by industry-wide benchmarks (usually well below true risk capital); and within a firm's divisions, risk capital and regulatory capital can diverge even if they match at the firm level (allocate the *greater* of the two per division).

The basic and detailed RAROC equations

BASIC RAROC

$$\text{RAROC} = \text{risk-adjusted return} / \text{risk-adjusted (economic) capital}$$

DETAILED RAROC (CAPITAL BUDGETING)

$$\text{RAROC} = [\text{expected revenues} - \text{costs} - \text{expected losses (EL)} - \text{taxes} + \text{return on economic capital} - \text{transfers}] / \text{economic capital}$$

WORKED EXAMPLE

\$1.5B loan portfolio, 7% pre-tax return, \$10M opex, funded by \$1.5B deposits at 5%, EL=0.5%/yr, UL=8% (\$120M economic capital), risk-free rate 1%, tax rate 25%.

Expected revenue=\$105M, interest expense=\$75M, EL=\$7.5M, return on econ. capital=\$1.2M.

$$\text{RAROC} = [(105 - 75 - 7.5 + 1.2) \times (1 - 0.25)] / 120 = 8.56\% \text{ after-tax.}$$

Risk capital $\approx$ the firm's one-year VaR at a high confidence level (usually $\geq 95\%$) — the buffer against *unexpected* loss specifically. Strategic risk capital = goodwill (excess purchase price over fair value of net assets) + burned-out capital (start-up spend at risk if a venture is abandoned, amortized over time).

Time horizon, default probability, confidence level — the three judgment calls

Time horizon: usually one year; the square-root-of-time rule scales 1-day VaR to annual, adjusted for "time to reduce" risk to a core level.

WORKED EXAMPLE

Daily VaR=80, core risk=60, 10 days to reduce (2/day), 252 business days: annualized VaR = $80 \times \sqrt{252} = 1,269.96$. Required risk capital $\approx 75.6\%$ of annualized VaR (blending the reduction path with the core-risk floor).

Default probability: point-in-time (PIT) for short-term pricing/expected loss; through-the-cycle (TTC) for economic capital, profitability, and strategic decisions — TTC produces lower economic-capital volatility since ratings change less often under a TTC lens. **Confidence level:** must match the firm's target credit rating (AA/AAA needs $>99.95\%$) — lowering it cuts required capital most dramatically for firms whose big losses are rare (op/credit/settlement risk-heavy books).

Hurdle rate and adjusted RAROC

HURDLE RATE (AFTER-TAX)

$$h_{AT} = \text{after-tax weighted average cost of equity capital (via CAPM)}$$

Decision rule: $\text{RAROC} > \text{hurdle rate} \rightarrow \text{accept (value creation)}$; $\text{RAROC} < \text{hurdle rate} \rightarrow \text{reject}$. Problem: this ignores that high-RAROC projects can still be high-risk, and low-RAROC projects can be low-risk and value-preserving — hence **adjusted RAROC (ARAROC)**, which nets out systematic risk via CAPM.

ADJUSTED RAROC DECISION RULE

$\text{ARAROC} > \text{risk-free rate} \rightarrow \text{accept}$ — $\text{ARAROC} < \text{risk-free rate} \rightarrow \text{reject}$

WORKED EXAMPLE

$\text{RAROC}=12\%$, $r_f=5\%$, market return=11%, $\beta=1.5$. $\text{ARAROC} = \text{RAROC} - \beta \times (\text{market return} - r_f) = 12\% - 1.5 \times 6\% = 3\%$. Since $3\% < 5\%$ (r_f), **reject** the project.

Diversification benefits — stand-alone, fully diversified, marginal capital

WORKED EXAMPLE

Activity A alone needs \$50, Activity B alone needs \$60, together they need \$90. Diversification benefit = $50+60-90 = \$20$, allocated pro-rata: A gets \$9.1, B gets \$10.9 → fully diversified capital: A=\$40.9, B=\$48.1.

Marginal capital: A's marginal = $90-60=\$30$; B's marginal = $90-50=\$40$ (marginal capital sums to less than total capital — this is normal and expected).

Use cases: stand-alone capital → incentive pay; fully diversified capital → solvency and minimum capital requirements; marginal capital → active portfolio/business-mix decisions. Caution: correlations can collapse toward ± 1 in a crisis, wiping out assumed diversification benefits exactly when you need them most.

RAROC best practices

Senior management buy-in and active promotion · clear communication/education for "buy-in" across management levels · ongoing consultation via a cross-functional metrics-review committee · centralized data quality control (RAROC team owns collection/computation/reporting; business units own data accuracy controls) · complement RAROC with a qualitative four-quadrant (return × earnings-quality) assessment · active capital management via quarterly limit requests reviewed collaboratively.

TRAP

A longer RAROC time horizon isn't automatically "better" — risk and return data over one year quickly becomes unreliable, which is exactly why one year remains the default despite the appeal of capturing a "full business cycle."

Range of Practices and Issues in Economic Capital Frameworks

The story

A challenges-and-limitations tour of everything economic capital modeling has to get right: choosing a risk measure, aggregating across risk types, validating the model, modeling credit dependency, handling counterparty credit risk, and pricing interest rate risk in the banking book. Mostly qualitative — treat it as a checklist reading.

Risk measure tradeoffs

Measure	Main weakness
Standard deviation	Not coherent (violates monotonicity); depends on distributional assumptions
VaR (most used)	Not coherent (violates subadditivity) — can distort internal capital allocation/limits
Expected shortfall	Harder to interpret; link to target credit rating unclear
Spectral/distorted measures	Rarely used, not intuitive — mainly academic

Five aggregation methodologies

Simple summation (ignores diversification) · constant diversification (summation minus a fixed %) · variance-covariance matrix (flexible but hard to estimate correlations, misses nonlinearity/skew) · copulas (hard to validate, hard to build the joint distribution) · full modeling/simulation (most demanding, can create false confidence). The variance-covariance approach is most common in practice despite its estimation difficulty — banks often use a deliberately conservative (upward-biased) correlation matrix.

Six qualitative + six quantitative model validation processes

Qualitative: use test, qualitative review, systems implementation testing, management oversight, data quality checks, assumption/sensitivity testing. Quantitative: input/parameter validation, model replication (rarely sufficient alone), benchmarking/hypothetical portfolios (only compares model-to-model, not to reality), backtesting (needs a quantifiable, comparable metric — not always available), P&L attribution (mostly used for market risk pricing models only), stress testing.

Dependency modeling, counterparty credit risk, IRRBB — the persistent challenges

Credit dependency modeling questions the ASRF/Gaussian-copula assumption itself: normal-distribution defaults, stable correlations over time, and whether PD-LGD correlation and LGD variability are adequately captured. For counterparty credit risk: market-risk-style simulation doesn't allow netting across counterparties the way portfolio VaR does, and exposure must be modeled over multiple future horizons, not one short window. For interest rate risk in the banking book (IRRBB): embedded optionality (mortgage prepayment on the asset side; the bank's rate-setting option and the depositor's withdrawal option on the liability side) makes cash flows genuinely indeterminate, requiring complex stochastic-path modeling.

BIS's 10 recommendations & economic capital's practical uses

10 BIS recommendations span: model use in capital adequacy assessment, senior management commitment, transparency/decision integration, risk identification, risk measures, risk aggregation, validation, credit dependency modeling, counterparty credit risk, and IRRBB.

Four practical application areas, each with constraints and opportunities: **credit portfolio management** (protects against risk deterioration) · **risk-based pricing** (maximizes profitability via minimum RAROC-implied rates) · **customer profitability analysis** (identifies unprofitable relationships to drop) · **management incentives** (motivates staff participation in the capital-allocation process, though studies show compensation is a minor actual driver of business-unit-level economic capital use).

Capital Planning at Large Bank Holding Companies

The story

Takes everything conceptual from R58 and shows how one real supervisory regime — the Federal Reserve's Capital Plan Rule — operationalizes it for large (\$50B+) U.S. bank holding companies. Read this as "R58's checklist, now with a named regulator and named process."

The Capital Plan Rule & the seven CAP principles

Applies to top-tier U.S. BHCs with \$50B+ consolidated assets, requiring a robust internal capital plan — the capital adequacy process (CAP). Seven principles span: risk management foundation, resource/loss estimation methods, capital adequacy, capital planning/internal controls policies, and governance oversight.

Key mechanics

Firm-wide risk identification must be comprehensive; models need independent, regular review/validation; the board must actively evaluate and approve the capital plan; a capital policy must define goals for issuance, usage, and distributions. Loss estimation should rest on sound theoretical/empirical grounds, blending quantitative and qualitative methods. Counterparty credit risk uses either a **probabilistic** approach (must show scenarios at least as severe as past observed events) or a **deterministic** approach (must span a wide range of stress scenarios).

Pre-provision net revenue (PPNR) projections must capture interrelationships among revenue, expense, and on/off-balance-sheet drivers within a scenario; net interest income methodology must incorporate current and projected balance sheet positions; noninterest income projections must reflect the stated scenario and business strategy.

THREAD

This is the micro-level implementation of exactly the macro-to-micro stress testing challenge R56 raised in the abstract — R59 is what that translation looks like inside one bank's actual capital planning process.

Capital Regulation Before the Global Financial Crisis

The story

The origin story of modern bank capital regulation, told through two accords: Basel I (1988) — the first risk-weighting system, credit risk only — and Basel II (2004/2007) — three pillars, sophisticated credit risk options, and (for the first time) operational risk capital. This reading has more testable formulas than almost any other reading in this book; budget serious drilling time.

Basel I: three capital ratios & risk-weighted assets

BASEL I'S THREE REQUIREMENTS

Total assets / capital < 20 (i.e. capital/assets > 5%) — Tier 1 / RWA > 4% — Total capital (Tier1+Tier2) / RWA > 8% (the "Cooke ratios")

Tier 1 (core): equity minus goodwill, noncumulative perpetual preferred. Tier 2 (supplementary, capped at 50% of total capital): cumulative perpetual preferred, certain 99-year debentures, subordinated debt >5yr original maturity.

RWA WORKED EXAMPLE

\$20M T-bills (0%) + \$20M insured mortgages (0%) + \$50M uninsured mortgages (50%) + \$150M corporate loans (100%) = 0+0+25+150 = **\$175M RWA**.

Off-balance-sheet items: credit equivalent amount (CEA)

CEA FOR DERIVATIVES (CURRENT EXPOSURE METHOD)

$CEA = \max(V, 0) + D \times L$ — V = current derivative value, D = add-on factor, L = principal

WORKED EXAMPLE

\$175M swap, 3yr remaining, current value \$2.5M, add-on factor 0.5% → $CEA = 2.5 + (0.005 \times 175) = \$3.375M$. RWA (OECD bank counterparty, 20% weight) = \$675,000; RWA (corporate counterparty, 100% weight) = \$3.375M. Required capital (corporate case) = $(175 + 3.375) \times 0.08 = \$14.27M$.

1995 netting amendment

NET REPLACEMENT RATIO (NRR)

$NRR = \text{current exposure with netting} / \text{current exposure without netting}$

WORKED EXAMPLE

Exposures +20, -7, +5 (millions) → without netting: exposure = 20+5=\$25M; with netting: 20-7+5=\$18M → $NRR = 18/25 = 0.72$.

1996 Amendment: market risk capital

MARKET RISK CAPITAL CHARGE

$\text{capital} = \max(\text{previous day VaR}, m_c \times \text{average 60-day VaR}) + \text{specific risk charge}$

10-day, 99% VaR; $m_c \geq 3$, set by backtesting 250 days of 99% 1-day VaR against actual losses:

Exceptions (of 250)	Multiplier
<5	3.00
5/6/7/8/9	3.40/3.50/3.65/3.75/3.85
>10	4.00

WORKED EXAMPLE

Previous-day VaR=\$10M, 60-day avg=\$8M, m=3 → capital = $0.08 \times [12.5 \times (3 \times \$8M)] = \$24M$.

Basel II: three pillars

Pillar 1 minimum capital (credit + market + operational risk, still 8% of RWA total). **Pillar 2** supervisory review (regulator discretion for local conditions, ICAAP required). **Pillar 3** market discipline (mandatory quantitative/qualitative disclosure).

Basel II credit risk: three approaches

Standardized (Basel-I-like, but ratings now matter — sovereign risk weights 0%–150%, banks/corporates 20%–150%) · Foundation IRB (bank supplies PD only; LGD/EAD/M are supervisory values — LGD=45% senior, 75% subordinated, M usually 2.5) · Advanced IRB (bank supplies PD, LGD, EAD, and M).

IRB CAPITAL (VASICEK/GORDY ONE-FACTOR GAUSSIAN COPULA)

$$\text{Capital}_i = \text{EAD}_i \times \text{LGD}_i \times [\text{WCDR}_i - \text{PD}_i] \times \text{MA}$$

$\text{WCDR} = \text{DR}_{99.9}$, the 99.9th-percentile default rate

MATURITY ADJUSTMENT & RWA

$$\text{MA} = [1 + (M-2.5) \times b] / (1 - 1.5 \times b) \quad \text{—} \quad \text{RWA} = 12.5 \times \text{Capital required}$$

WORKED EXAMPLE

\$150M loan to A-rated corp, PD=0.1%, LGD=50%, $\text{DR}_{99.9}=3.4\%$, M=2.5yr. Capital = $150 \times 0.5 \times (0.034 - 0.001) = \$2.475M$ → RWA = $12.5 \times 2.475 = \$49.19M$ (vs. \$150M flat under Basel I — IRB substantially lowers RWA for higher-rated corporate exposures).

Retail exposures merge foundation/advanced IRB (bank supplies PD, EAD, LGD directly) and drop the maturity adjustment entirely: capital = $\text{EAD} \times \text{LGD} \times (\text{DR}_{99.9} - \text{PD})$.

Basel II operational risk: three approaches

Approach	Method
Basic indicator (BIA)	15% × 3-yr average annual gross income (negative-income years excluded from the average)
Standardized	Same idea, different multiplier per business line (e.g. 12% retail, 15% commercial, 18% payments/settlement)
Advanced measurement approach (AMA)	Internal one-year, 99.9% VaR model; can reflect insurance mitigation

BIA WORKED EXAMPLE

Gross income \$20B (Yr1), -\$2B (Yr2, excluded), \$12B (Yr3) → capital = $(20+12)/2 \times 0.15 = \$2.4B$.

Solvency II (insurance parallel to Basel II)

SCR (solvency capital requirement) sits above the MCR (minimum capital requirement) floor. Two approaches: standardized (average-firm risk profile) or internal models (one-year VaR at 99.5% — note: lower confidence than Basel II's 99.9% for banks). Three internal-model tests: statistical quality, calibration, use test.

TRAP

Basel II's IRB confidence level (99.9%) and Solvency II's internal model confidence level (99.5%) are frequently swapped by mistake on the exam — banks get the stricter 99.9%, insurers get 99.5%.

Solvency, Liquidity, and Other Regulation After the Global Financial Crisis

The story

The 2007–09 crisis exposed exactly where Basel II fell short, and this reading is the point-by-point fix: Basel 2.5 patches market risk capital (stressed VaR, incremental risk charge, comprehensive risk charge), Basel III tightens capital definitions and adds buffers, and a new liquidity regime (LCR, NSFR) addresses the risk that actually killed banks in 2008 — not insolvency, but liquidity. Closes with CoCos and the broader Dodd-Frank reform wave.

Basel 2.5: three fixes to market risk capital

TOTAL MARKET RISK CAPITAL CHARGE (WITH STRESSED VAR)

$$\text{capital} = \text{VaR} + \text{stressed VaR component} = \text{VaR}_{\text{previous day}} + (m_s \times \text{average stressed VaR, 60-day})$$

WORKED EXAMPLE

Previous-day VaR=\$15.6M, avg VaR(60d)=\$4.8M, previous-day stressed VaR=\$17.7M, avg stressed VaR=\$18.4M, $m_t=m_s=3 \rightarrow \text{total} = 15.6 + (18.4 \times 3) = \70.8M (roughly double the pre-2.5 capital charge, by design).

Stressed VaR uses a 250-day window from the worst one-year period in the bank's own most recent seven years (not a fixed 2008 assumption — the stress window differs bank to bank). **Incremental risk charge (IRC)** (evolved from the 2005 incremental default risk charge) requires a 99.9%, one-year VaR for credit-sensitive trading-book instruments, using a **constant level of risk** assumption — the bank is assumed to rebalance back to the original position at each instrument's estimated liquidity horizon (minimum 3 months), booking small losses on downgrades rather than riding out full defaults. **Comprehensive risk (CR) charge** replaces the specific risk charge and IRC for correlation-sensitive instruments (ABS, CDOs) — unrated or below-BB- tranches require dollar-for-dollar (100%) capital.

Basel III capital tiers & ratios

Tier 1 equity (common equity + retained earnings, minus goodwill/intangibles/DTAs/DVA effects) + additional Tier 1 (noncumulative perpetual preferred, contingent-convertible debt) + Tier 2 (subordinated debt ≥5yr, some preferred, limited loan loss reserves). Tier 3 eliminated entirely.

BASEL III MINIMUM RATIOS (NORMAL TIMES)

$$\text{Tier 1 equity} / \text{RWA} \geq 4.5\% \quad \text{Total Tier 1} / \text{RWA} \geq 6\% \quad \text{Total capital} / \text{RWA} \geq 8\%$$

Capital buffers

WITH CAPITAL CONSERVATION BUFFER (CCB, 2.5% OF RWA, TIER 1 EQUITY)

$$\text{Tier 1 equity} \geq 7.0\% \quad \text{Total Tier 1} \geq 8.5\% \quad \text{Total capital} \geq 10.5\%$$

Dividends are constrained as the buffer is drawn down (e.g., at a 6% Tier 1 equity ratio, a bank must retain ≥60% of earnings). **Countercyclical buffer (CCyB)**: 0–2.5% of RWA, at national discretion, weighted across jurisdictions for international banks — rationale: dampen credit overheating, and it's cheaper to raise capital in good times than bad. **G-SIB buffer**: additional 1%–3.5% (5 tiers) for globally systemically important banks, plus (since 2017) a G-SIB leverage ratio buffer.

Liquidity ratios — the crisis's real lesson

LEVERAGE RATIO

$$\text{capital} / \text{total exposure} \geq 3\% \quad (\text{unweighted, includes some off-balance-sheet items})$$

LIQUIDITY COVERAGE RATIO (LCR)

$$\text{high-quality liquid assets (HQLA)} / \text{net cash outflows over 30 days} \geq 100\%$$

NET STABLE FUNDING RATIO (NSFR)

available stable funding (ASF) / required stable funding (RSF) $\geq 100\%$

WORKED EXAMPLE

HQLA (cash+central bank reserves+Treasures, 0% haircut) = \$30 (loans/fixed assets get 0% credit). Net 30-day outflow = $(100 \times 5\% \text{ retail}) + (75 \times 20\% \text{ wholesale}) + (20 \times 0\% \text{ capital}) = \20 . **LCR = $30/20 = 150\%$ ✓**.

ASF = $(100 \times 0.9) + (75 \times 0.5) + (2 \times 1.0) + (18 \times 1.0) = \147.50 . RSF (across loan/asset buckets by factor) = \$137.00. **NSFR = $147.50/137.00 = 107.66\%$ ✓** — both ratios clear the 100% bar.

LCR haircuts: central bank/government securities 0%, corporate debt/equity 50%, individual mortgages excluded entirely. HQLA must survive a 30-day severe stress: a 3-notch downgrade, deposit runs, total wholesale funding loss, collateral devaluation, and credit line drawdowns.

Contingent convertible bonds (CoCos)

Unlike ordinary convertibles (bondholder chooses to convert), CoCos convert to equity *automatically* on a trigger — typically a Tier 1 equity/RWA threshold breach, supervisor judgment of insolvency risk, or a market-cap/ assets ratio floor. Motivation: debt in normal times (doesn't drag ROE), automatic equity cushion in stress (private capital absorbs losses instead of a government bailout).

Dodd-Frank and the broader reform wave

FSOC (systemic risk monitoring) · SIFIs must maintain living wills · compensation reform (discourage short-termist risk-taking, shareholder say-on-pay) · OTC derivatives moved to central clearing/ exchanges, SEFs mandated · increased rating-agency transparency (Office of Credit Ratings) · Consumer Financial Protection Bureau (CFPB) · ability-to-repay mortgage requirements · minimum 5% securitization retention ("skin in the game") · Volcker Rule (bars proprietary trading by deposit-taking banks).

TRAP

The capital conservation buffer is a mandatory requirement (built up in normal times, drawn down in stress) — it is *not* discretionary the way the countercyclical buffer is left to national regulators. Don't conflate the two buffers' governance.

High-Level Summary of Basel III Reforms

The story

A deliberately brief bridge reading — the December 2017 reforms (effective 2022) that finalize Basel III, setting up R63's detailed SMA formula. GARP explicitly says: don't get lost in the mechanics here, know the goals and the big-picture shifts.

Four goals of the December 2017 reforms

Expand the standardized approach's robustness/sensitivity (credit, CVA, operational risk) · restrict internal model approaches for the same three risk types · add a G-SIB leverage ratio buffer · create a more risk-sensitive output floor.

What changed, by risk type

Risk type	Change
Standardized credit risk	More granular risk weights (e.g., mortgage risk weight now depends on LTV); reduced reliance on external ratings
IRB credit risk	A-IRB removed for large/mid corporates and financial institutions (forced to F-IRB); input floors added for PD/LGD/EAD
CVA risk	IRB approach removed entirely; must use standardized (SA-CVA) or basic (BA-CVA) approach
Operational risk	AMA and all prior SAs replaced by one single SA (the SMA, detailed in R63) driven by income + historical losses
Leverage ratio	New G-SIB leverage buffer; refined derivatives/off-balance-sheet exposure measure

The output floor

OUTPUT FLOOR RULE

$$RWA = \max(\text{RWA under bank's approved approach}, 72.5\% \times \text{RWA under the standardized approach})$$

Purpose: stop large banks from using internal models to shrink capital requirements far below what the standardized approach would demand. Standardized building blocks used to compute the floor: SA for credit risk, SA-CCR for derivatives counterparty risk, SA-CVA/BA-CVA (or 100% of counterparty credit risk capital) for CVA, SEC-ERBA/SEC-SA/1,250% weight for securitization, and the plain SA for both market risk and operational risk.

TRAP

IRB approaches are explicitly barred from the output-floor calculation for *every* risk type listed — including CVA risk, where the IRB option was removed from the framework altogether, not just from the floor calculation.

Basel III: Finalizing Post-Crisis Reforms

The story

This closes the entire book by giving you the actual formula behind "the SA for operational risk" that R60's BIA/SA/AMA history and R62's reform summary were both building toward: the **Standardized Measurement Approach (SMA)**. It combines a financial-statement proxy (business indicator) with the bank's own loss history (internal loss multiplier) into one non-model-based capital number — deliberately simpler and more comparable across banks than the AMA it replaced.

The Business Indicator (BI) — three components, 3-year averages

BI

$$BI = ILDC + SC + FC$$

Component	Captures
ILDC (interest, lease, dividend component)	Net interest income (capped relative to interest-earning assets) + dividend income
SC (services component)	max(fee income, fee expense) + max(other operating income, other operating expense)
FC (financial component)	Absolute value of net trading book P&L + absolute value of net banking book P&L

BI buckets and the business indicator component (BIC)

BIC — MARGINAL COEFFICIENTS BY BUCKET (ILLUSTRATIVE STRUCTURE)

Bucket 1 ($BI \leq €1bn$): $BIC = 12\% \times BI$
 Bucket 2 ($€1bn < BI \leq €30bn$): marginal coefficient rises to 15%
 Bucket 3 ($BI > €30bn$): marginal coefficient rises to 18%

WORKED EXAMPLE

BI = €40 billion (bucket 3) — BIC computed by applying the bucket 1 rate to the first €1bn, the bucket 2 rate to the next €29bn, and the bucket 3 rate to the remainder above €30bn, then summing (a marginal, tax-bracket-style calculation).

Internal loss multiplier (ILM) — where a bank's own history enters

ILM

$$ILM = \ln[e + (\text{Loss Component} / BIC)^{0.8} - 1]$$

e = Euler's number (≈ 2.71828), not e to a power

Loss Component = $15 \times$ average annual operational risk losses over the last 10 years (or the bank's actual historical average, scaled). If a bank's loss experience is exactly industry-average, Loss Component = BIC $\rightarrow$ ILM = 1 $\rightarrow$ capital = BIC unchanged. Worse-than-average loss history $\rightarrow$ Loss Component > BIC $\rightarrow$ ILM > 1 $\rightarrow$ capital requirement rises above BIC; better-than-average $\rightarrow$ ILM < 1 $\rightarrow$ capital falls below BIC.

Data requirement: ideally 10 years of quality loss data; 5 years permitted during transition; with less than 5 years, ILM is skipped entirely and capital = BIC alone.

Full capital requirement, by bucket

STANDARDIZED APPROACH CAPITAL

Bucket 1: capital = BIC (ILM not used) — Buckets 2 & 3: capital = BIC $\times$ ILM

WORKED EXAMPLE

PS Bank, BI = €18.48 million (bucket 1, since BI ≤ €1 billion range applies at this small scale) → capital = $0.12 \times 18.48 = \text{€2.22 million}$ (BIC only — bucket 1 banks never apply the ILM).

Consolidation rules: fully consolidated BI at the group level (netting intragroup income/expenses); at subsidiary/subconsolidated level, use that entity's own BI. If a subsidiary crosses into bucket 2/3 territory, it must incorporate its *own* loss experience (not the group's) — and if it doesn't meet the qualitative standards required to use the loss component, capital defaults to 100% of BIC.

SMA vs. AMA — why the framework changed

AMA (introduced 2006 under Basel II) was principles-based and flexible, hoping best practices would emerge and converge over time — instead it produced poor cross-bank comparability and overly complex modeling with no convergence. The SMA replaces it with a single, non-model-based formula combining financial-statement data with bank-specific loss experience — less flexible, but comparable and simple by design.

Operational loss data criteria (general + specific)

General: documented identification/collection/treatment processes; track gross loss, occurrence/ discovery/accounting dates, recoveries, and loss drivers; independent accuracy review; credit-RWA-linked losses excluded, market-risk-linked losses included; categorize to Basel Level 1 categories; 10-year observation window (5-year exception on first adoption); comprehensive across all geographies/subsystems; gross loss threshold €20,000 on first adoption, rising to €100,000 allowed for bucket 2/3 banks thereafter.

Specific: a documented policy defines what enters the SMA loss dataset; gross loss, insurance recoveries, and non-insurance recoveries tracked separately (dataset uses losses net of insurance recoveries only). Gross loss *includes*: external expenses (legal, advisory, vendor, repair/replacement) tied to the event, settlements/ impairments/write-downs/direct income-statement charges, reserves/provisions booked, material pending losses in suspense accounts, and material timing losses spanning multiple accounting periods. Gross loss *excludes*: post-event improvement/upgrade costs, insurance premiums, and general PP&E maintenance contract costs. The date of accounting is the only date usable for building the loss dataset (for legal losses, the date the legal reserve is first booked).

TRAP

General maintenance contracts on property, plant & equipment are explicitly *excluded* from the SMA gross loss calculation — even though they might feel "operational" in a loose sense, they're ordinary running costs, not operational risk losses. This exact exclusion is a favorite distractor on the exam.

Cross-Reading Synthesis: The Five Threads of Book 3

Operational Risk reads more like a collection of domains than a single connected model — but five ideas recur across sessions and are worth tracking as threads, not isolated facts.

1. Three lines of defense, everywhere

Introduced in R41 for general operational risk governance, then reapplied nearly verbatim to cyber risk (R47), AML/CFT (R49), and ERM generally (R46). Once you know the Line 1 (owns risk) / Line 2 (independent oversight, no self-review) / Line 3 (fully independent audit) structure, you can answer governance questions in almost any later reading without relearning the framework.

2. $EL = PD \times LGD \times EAD$ reappears as the capital-formula backbone

First fully quantified in Credit Risk Book 2, this exact structure resurfaces in R60's IRB capital formula ($Capital = EAD \times LGD \times [WCDR - PD]$), and its logic — expected loss covered by pricing, unexpected loss covered by capital — is exactly the distinction R57's RAROC numerator (subtracting EL) and denominator (economic capital for UL) are built around.

3. The Basel arc: I → II → 2.5 → III → the 2017 finalization

R60–63 form a single connected timeline, not four separate topics: Basel I invents risk-weighting (credit only) → Basel II adds sophistication (IRB, three pillars) and operational risk capital for the first time → the 2007–09 crisis exposes Basel II's gaps, patched by Basel 2.5 (stressed VaR, IRC, CR charge) and Basel III (tighter capital definitions, buffers, liquidity ratios) → the 2017 reforms restrict internal models and finalize a single standardized operational risk formula (the SMA in R63). Study these four readings as one continuous story, not four separate rulebooks.

4. VaR-at-99.9%-over-one-year is the recurring capital yardstick

Basel II's IRB approach (R60), the AMA operational risk approach (R60), and general economic capital approaches (R58) all converge on the same statistical target: a one-year, 99.9% confidence loss threshold, with capital covering the gap between that threshold and expected loss. Solvency II is the one deliberate exception (99.5%, not 99.9%) — a frequently tested contrast.

5. Loss data mechanics recur from R43 to R63

The four key dates (occurrence, discovery, reporting, accounting), the €20,000 minimum threshold, boundary event treatment, and the internal/external data-blending methods (scaling, cut-off mix, filtering) are introduced in R43's LDA discussion and then reappear almost unchanged as the SMA's data-quality criteria in R63. If you know R43's loss-data rules well, R63's "general and specific criteria" section is mostly recognition, not new learning.

Formula-dense vs. conceptual readings — where to spend practice-question reps

Heaviest formula load (drill hardest): R43 (fault tree, LDA), R57 (RAROC), R60 (Basel I/II ratios, IRB capital), R61 (Basel 2.5/III ratios, LCR/NSFR), R63 (SMA formula).

Mostly conceptual (fast review, spot-check with questions): R40, R41, R42, R44, R45, R46, R47, R48, R49, R50, R51, R52, R53, R54, R55, R56, R58, R59, R62.

Complete Formula Index

R43 — Risk Measurement

FTA: joint P = product of independent control-failure probabilities

LDA: frequency (Poisson) $\otimes$ severity (lognormal) $\rightarrow$ 99.9th pctile capital

R57 — RAROC

economic capital = risk capital + strategic risk capital

RAROC = risk-adj. return / risk-adj. (economic) capital

RAROC (detailed) = [rev-costs-EL-tax+return on capital-transfers]/econ. capital

Annualized VaR = daily VaR $\times \sqrt{252}$

ARAROC = RAROC - β (market return - r_f)

Diversification benefit = Σ (stand-alone) - fully diversified

Marginal capital = total capital - capital ex this activity

R60 — Capital Regulation Pre-GFC

Basel I: Tier1/RWA > 4%, Total/RWA > 8%

CEA = $\max(V, 0) + D \times L$

NRR = exposure w/ netting / exposure w/o netting

Mkt risk capital = $\max(\text{VaR}, m_c \times \text{avg 60d VaR}) + \text{SRC}$

$\text{Capital}_i = \text{EAD} \times \text{LGD} \times (\text{WCDR} - \text{PD}) \times \text{MA (IRB)}$

RWA = 12.5 $\times$ capital required

BIA op risk capital = 15% $\times$ 3yr avg gross income

R61 — Post-GFC Regulation

Total mkt risk capital = VaR + $m_s \times \text{avg stressed VaR}$

Basel III mins: Tier1 eq $\geq 4.5\%$, Tier1 $\geq 6\%$, Total $\geq 8\%$ of RWA

With CCB: Tier1 eq $\geq 7\%$, Tier1 $\geq 8.5\%$, Total $\geq 10.5\%$

Leverage ratio = capital/total exposure $\geq 3\%$

LCR = HQLA / 30-day net cash outflow $\geq 100\%$

NSFR = ASF / RSF $\geq 100\%$

R62 — Basel III Reforms Summary

Output floor: RWA = $\max(\text{approved approach RWA}, 72.5\% \times \text{SA RWA})$

R63 — SMA (Finalized Basel III)

BI = ILDC + SC + FC

BIC = bucket marginal coefficient $\times$ BI (tiered, tax-bracket style)

ILM = $\ln[e + (\text{Loss Component}/\text{BIC})^{0.8} - 1]$

Bucket 1: capital = BIC $\cdot$ Buckets 2&3: capital = BIC $\times$ ILM